



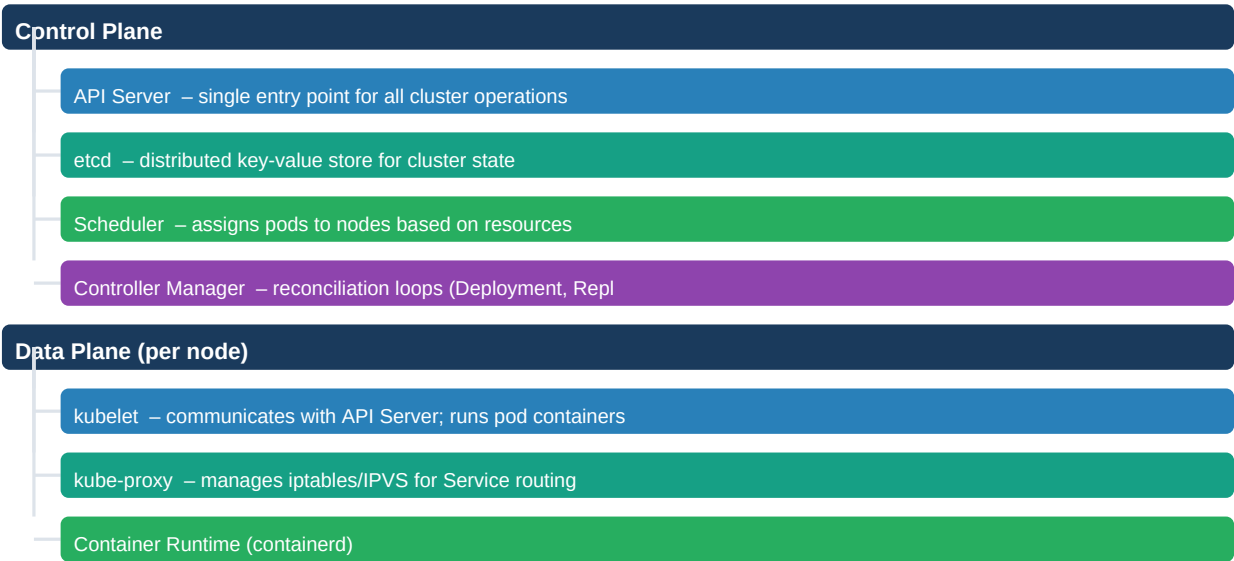
Kubernetes

Workloads · Helm · HPA · Ingress · GKE/EKS

Overview

K8s manifests + Helm chart for ShopVerse. HPA on CPU+custom metrics, PodDisruptionBudget for zero-downtime rollouts, cert-manager TLS, and Fabric8 Java client for dynamic Job creation.

Kubernetes Architecture



Key Manifest Files

File	Kind	Key Config
k8s/deployment.yaml	Deployment	replicas:3, RollingUpdate, readiness/liveness → /actuator/health
k8s/service.yaml	ClusterIP + LoadBalancer	Internal + external access; DNS shopverse-postgres.default.svc
k8s/ingress.yaml	Ingress (Nginx)	TLS via cert-manager ClusterIssuer; routes /api → shopverse-web:8080
k8s/hpa.yaml	HorizontalPodAutoscaler	min:2, max:10; CPU>70% OR http_requests_per_second>100
k8s/pdb.yaml	PodDisruptionBudget	minAvailable:1 – zero downtime during rolling updates
k8s/configmap.yaml	ConfigMap	datasource.url, redis.host, kafka.bootstrap-servers
k8s/secret.yaml	Secret	DB password, JWT_SECRET, STRIPE_API_KEY (base64)
k8s/cronjob.yaml	CronJob	Monthly partition creation job

Helm Chart Structure

k8s/helm/shopverse/

Chart.yaml – name, version, appVersion

values.yaml – default values (image tag, replicas, resource

values-gke.yaml / values-eks.yaml – cloud overrides

templates/deployment.yaml – uses {{ .Values.* }}

templates/service.yaml

templates/ingress.yaml

templates/hpa.yaml / pdb.yaml / configmap.yaml / secre